Core revision exercises: Number

Worksheet 1: Reviewing number concepts

1 List the first five multiples of:

a 9 **b** 17

2 What is the HCF of:

a 36 and 144 **b** 32 and 60?

3 What is the LCM of:

a 12 and 15 **b** 12 and 18?

4 List all the prime numbers between:

a 20 and 50 **b** 55 and 100

5 Write as a product of prime factors:

a 200 **b** 189

6 Without doing any calculation, state whether the following are true or false.

a $4.87 \times 6 = 292.2$ **b** $40.76 \div 1 = 40.76$

c $0.12^2 < 0.1$ **d** $\sqrt{0.25} = 0.05$

e $31.6 \times 2.894 \neq 3.16 \times 28.94$ **f** $6.8 \geqslant 0$

g $18.62 \leqslant 9 \times 2.1$

7 Calculate, giving your answers correct to two decimal places where necessary.

a $9 + 7.047$ **b** $12.88 + 5.37$

c $12.72 - 9.004$ **d** $\dfrac{202.25}{35.6}$

e $\dfrac{12.17 \times 3.142}{2.155}$ **f** $\dfrac{14.27 + 19.29}{12.6 - 4.03}$

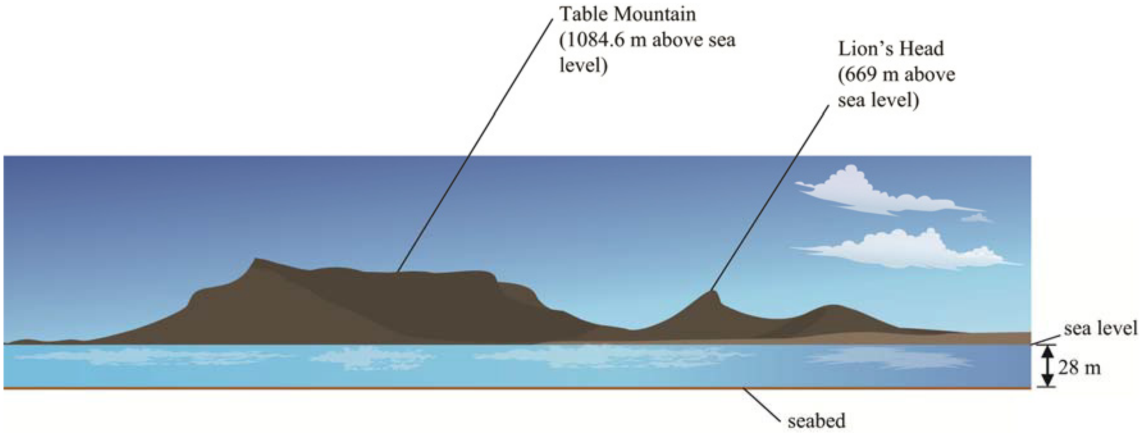
8 Arrange these temperatures in order from coldest to hottest:

a 8°C, 2°C, 4°C, 0°C, −2°C **b** −4°C, 7°C, 0°C, 19°C , 29°C, −12°C

9 The temperature in Hokkaido in Japan was −8°C at 00:00 one night.

By 09:00 the temperature had risen 12 degrees. What was the temperature at 9 am?

10 Study the diagram.



a Express the depth of the sea as a directed number. HINT: sea level is taken to be 0.

b Calculate the difference in height between the top of Table Mountain and the seabed.

c If sea levels rose 1.85 m due to global warming, how would it affect the altitude of Lion's Head?

11 Evaluate. Give your answers correct to 2 decimal places if necessary.

a $\sqrt{5}+\sqrt{13}$ **b** $\sqrt[3]{8}-\sqrt{1}$ **c** $2\sqrt{62}+3\sqrt{81}$

d $4^2\times\sqrt[5]{7776}$ **e** $2^4+\sqrt[4]{390\ 625}$ **f** $\dfrac{149}{11-\sqrt{2}}$

g $\dfrac{15}{48+2\sqrt{7}}$ **h** $\sqrt[3]{1728}-\sqrt{16129}$

12 Calculate. Give your answers correct to 3 decimal places if necessary.

a $14\times67+12\times42$ **b** $\dfrac{77}{14}\times\dfrac{29}{11}$

c $(427+842-374)\div(289+76)$ **d** $(0.467)^2\times\sqrt{900}$

e $\left(\sqrt{8}\right)^2\times(2.127)^3$ **f** $\left(\dfrac{5}{6}\right)^2+\left(\sqrt{144}\right)^3$

g $(1.5)^2\times\sqrt[3]{81}+\sqrt{15}$ **h** $\sqrt[3]{205379}-6\left(\sqrt{343}\right)^2$

i $(687-205+379)\div(1.4\times209)-\sqrt{121}$

13 a Which two numbers have a sum of 94 and a product of 2013?

b Which two numbers have a difference of 19 and a product of 1170?

14 At a school swimming gala, a student stands on a 9 m high diving board and dives 15 m to touch the bottom of the pool.

How deep is the pool?

15 A married couple have four children. Each of their children marries and has three children. Assuming no one dies or gets divorced, how many people are now in the extended family?

Core revision exercises: Algebra

Worksheet 2: Making sense of algebra

1 Shamiel has 6 pieces of rope labelled *a*–*e*. The first piece, *a*, is *x* metres long.

Write an expression in *x* to describe the length of the other pieces using the following information:

a b is 6 m shorter than a.

b c is half as long as a.

c d is 2.5 m longer than a.

d e is a third of the length of a.

e f is twice as long as a.

2 Simplify.

a $4a-2a+6b$

b $4x\times 8y$

c $\frac{x}{3}\times\frac{2x}{3}$

d $2x^2y-6xy+xy^2+2xy-y^2x$

e $\frac{12xy^2}{7z}\times\frac{21z^2}{4xy}$

f $8x-2y-3x-7y$

g $\frac{12x}{10}$

h $\frac{25x^3}{50x^2y^2}$

3 Given that $a=2$, and $b=5$ and $c=8$, find the value of:

a $3(a+b)$

b a^2b

c $2a-b$

d $ab+bc$

e c^2-c

f $a+\frac{c}{a}$

g $\frac{a}{2}+\frac{c}{4}$

h ab^2+a^2b

i $c-3a$

j $ab-c$

k $\frac{a}{c}+\frac{a}{2b}$

l $\frac{2(a+c)}{b}$

m $\frac{c-a}{c-b}$

n $\frac{1}{2}a^3bc$

o $\sqrt[3]{c}$

4 Expand and simplify.

a $4(x+3)$

b $5(x-2)$

c $2x(x+4)$

d $3x(x-2)$

e $7x(x+y)$

f $3(x+5)-7$

g $4(x-2)+2x$

h $4(x-y)-2x$

i $4(x-3)+2(x+7)$

j $4x(x-2)+y(2y+3)$

k $6(x-7)+3(2-x)$

l $3x(x^2+5)+2x^2(2x-4)$

5 Write each of the following as a product of prime numbers.

a 52

b 26

c 36

d 144

e 32

f 81

g 1024

h 2450

6 Explain why it is not possible to write the number 41 as a product of prime numbers.

7 Simplify.

a $x^8 \div x^2$ **b** $y^8 \div y^8$ **c** $\left(x^2\right)^3$

d $\left(x^5\right)^3$ **e** $x^0 \times 2$ **f** $x^0 + y^0$

g $\frac{x^{10}}{x^{12}}$ **h** $2\left(x^2\right)^4$ **i** $6x^2 \times 4xy^2$

j $2x^2y^2 \times xy^2$ **k** $2\left(x^2\right)^4$ **l** $\left(4x^3\right)^3$

m $\frac{120x^2}{5x}$ **n** $\frac{12x^2y^2}{9x^3y^3}$ **o** $\left(4x^4\right)^3$

p $2a^{-1}$ **q** $\left(x^{-2}\right)^{-2}$ **r** $(2x)^{-2}$

s $2\left(x^2y\right)^{-1}$ **t** $\left(xy^2z\right)^{-2}$

8 Evaluate the following, rounding your answers to 3 significant figures where appropriate.

a $9^{\frac{1}{3}}$ **b** $41^{\frac{2}{5}}$ **c** $4^{\frac{5}{3}}$

d $216^{\frac{1}{3}}$ **e** $256^{0.5}$

9 Simplify.

a $x^{\frac{1}{2}} \times x^{\frac{1}{2}}$ **b** $x^{\frac{1}{5}} \times x^{\frac{2}{3}}$ **c** $\left(\frac{x^2}{x^9}\right)^{\frac{1}{3}}$

d $\left(\frac{x^2}{y^3}\right)^{\frac{5}{6}}$

10 Find the error in each of these simplifications and rewrite each one correctly.

a $4(x+3) = 4x+3$

b $4(x-2)+2(x+4) = 6x-4$

c $3(x-2)+5(x+1) = 8x-5$

d $2(x+3)+3(x+4) = 6x+18$

11 A school tuckshop sells w bottles of water and c bottles of cool drink each day for d days. Explain in your own words what each of these expression means.

a $(w+c)$

b $(w-c)$

c $(c-w)$

d dw

e $d(w+c)$

f $d(w+c)-8w$

Core revision exercises: Shape, space and measures

Worksheet 3: Lines, angles and shapes

1 Complete the table.

Name of polygon	Number of sides	Sum of angles	Size of one angle when the polygon is regular
Triangle	3	180°	60°
	4		
	5		
	6		
	8		
	10		

2 Calculate the size of one interior angle in a regular 18-sided polygon.

3 Construct triangle XYZ with $XY = 40$ mm, $YZ = 50$ mm and $XZ = 70$ mm.

4 Draw accurate sketches to show:

a an angle with a complement of 42°

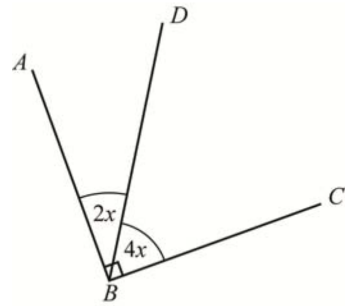
b an angle with a supplement of 42°

c an angle equal to 239°

d the pairs of equal angles formed when a transversal cuts two parallel lines.

5 Find the value of x in each diagram. Show all your working and give reasons for any statements.

a



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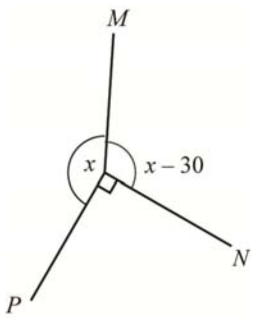

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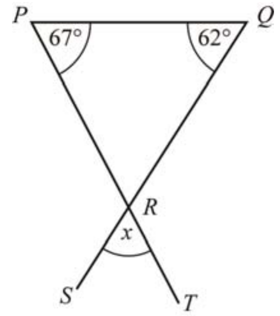

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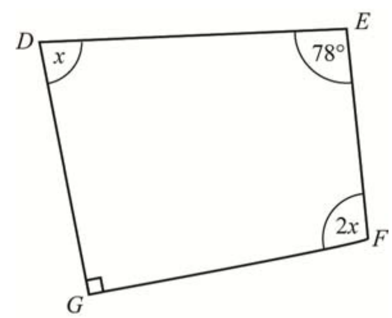

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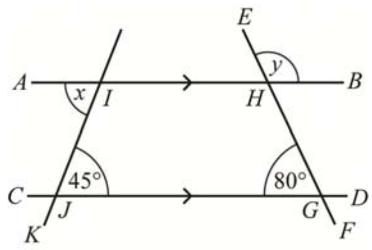

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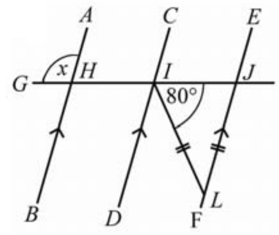

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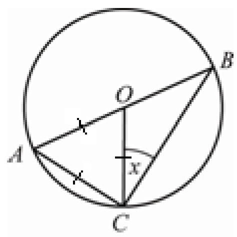

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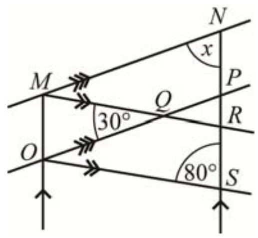
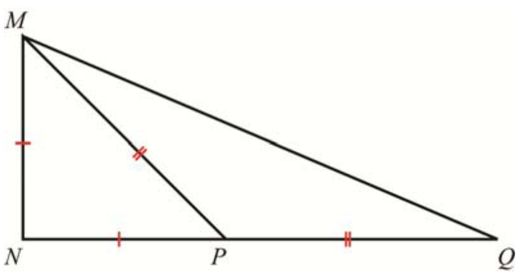


6 Prove that angle $NMQ = 3 \times$ angle MQN



Core revision exercises: Data handling

Worksheet 4: Collecting, organising and displaying data

1 A researcher asked the students in a school how many fizzy drinks they consumed during one particular week. These are her results:

8	19	25	25	0	0	0	2	28	25
13	19	13	18	15	0	0	10	9	24
23	0	0	33	32	0	0	29	0	4
5	0	16	15	3	6	0	33	23	0
30	13	22	21	6	8	9	8	12	20

a Is this discrete or continuous data?

b Organise the data into a grouped frequency table. Use a class interval of 0–4, 5–9 and so on.

c Draw a suitable graph to display these results.

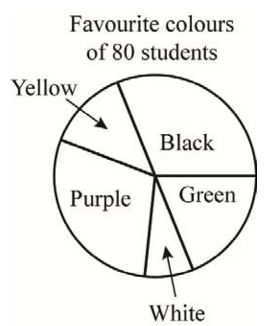
2 This pie chart shows the colours that 80 students selected as their favourite from a 5-colour chart.

a Which colour is most popular?

b Which colour is least popular?

c What percentage of the students chose purple as their favourite colour?

d How many students chose black as their favourite colour?



3 Here are the ages (to the nearest whole year) of the oldest grandparent of 40 six-year-old girls:

63	67	52	90	75	99	80	51	64	51
51	53	79	61	60	72	75	67	62	70
89	65	55	93	56	65	52	62	65	53
99	63	75	99	53	58	71	50	55	51

a Draw an ordered stem-and-leaf diagram to organise the data.

b Draw a bar graph to show this data.

4 The pulse rates (in beats per minute) of ten students were measured before and after they exercised for 10 minutes.

Before	80	55	73	64	55	50	69	74	67	69
After	103	84	120	113	121	115	95	115	97	98

a Draw an ordered back-to-back stem-and-leaf diagram to display the data.

b What conclusion can you draw from the diagram?

5 The favourite subjects of a group of students in a school in Dhaka is shown in the table.

Subject	**Girls**	**Boys**
Mathematics	34	33
English	45	40
Biology	29	31
ICT	40	48

a Draw a double bar graph to show this data.

b How many girls chose mathematics?

c How many boys chose ICT?

d Which is the favourite subject among the girls?

e Which is the least favourite subject amongst the boys?

6 Study this graph carefully.

a What type of graph is this?

b What does each symbol represent?

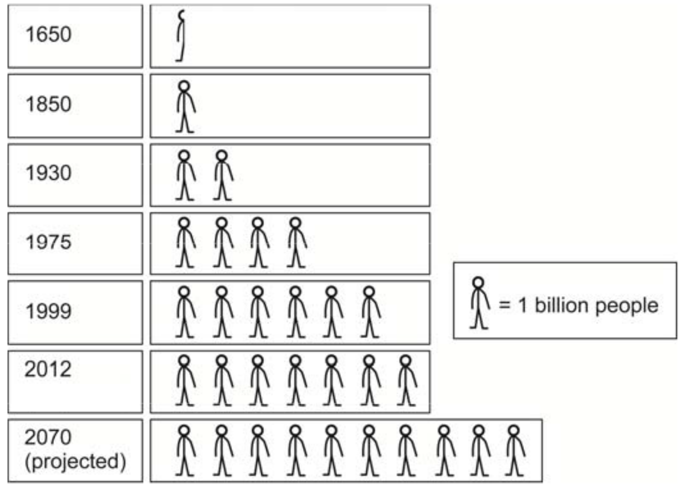
c What was the population of the world in 1650?

d How long did it take the population to double after 1650?

e When did the world's population reach 7 billion?

f The United Nations predicts that the world's population will reach 9.2 billion in 2050. How would you show this on the graph?

g Redraw this data as a line graph.



7 This table shows the approximate percentage of the world's population living on each continent.

Africa	Asia	Europe	North America	South America	Oceania
16.6	59.6	9.9	4.8	8.5	0.6

a Draw a pie chart to display this data.

b How else could you display this data?

Core revision exercises: Number

Worksheet 5: Fractions and standard form

1 Round each of the following numbers to the accuracy shown in brackets.

a 15.638 (1dp) **b** 383 452 345 (3sf)

c 0.000 034 556 (2sf) **d** 0.999 98 (3dp)

e 32 453 (nearest 10) **f** 0.123 45 (nearest thousandth)

g 125 (nearest 10)

2 Write each of the following as an ordinary number.

a 5.3×10^{6} **b** 9.56×10^{12} **c** 1.08×10^{8}

d 8.75×10^{9} **e** 5.3×10^{-3} **f** 2.08×10^{-6}

g 9.1×10^{-7} **h** 2.145×10^{-8}

3 Write each of the following numbers in standard form.

a 65 000 000 **b** 34 800 000 000 **c** 0.000 19

d 0.000 000 127 **e** 12 000 **f** 0.0056

g 27 500 000 **h** 0.000 000 000 000 345

4 Simplify. Leave your answers in standard form.

a $(4\times10^{2})\times(2\times10^{5})$ **b** $(7\times10^{-3})\times(4\times10^{-1})$

c $(3.45\times10^{4})\times(3\times10^{-2})$ **d** $(4.6\times10^{-2})\times(7.2\times10^{5})$

e $(9.5\times10^{18})\times(4.7\times10^{9})$ **f** $(0.76\times10^{-4})\times(0.04\times10^{-3})$

5 Complete this table.

Fraction	Decimal	Percentage
$\frac{1}{3}$		
	0.77	
		35
	1.4	
	0.125	
		37.5
		95

Fraction	Decimal	Percentage
$\frac{7}{12}$		
$1\frac{4}{5}$		
$\frac{9}{20}$		
		60
		75
	0.68	
	0.33	

6 Evaluate:

a $\frac{2}{5}+\frac{2}{7}$ **b** $\frac{5}{8}+\frac{3}{4}$ **c** $3\frac{3}{8}+4\frac{7}{10}$

d $3\frac{2}{9}+2\frac{2}{5}$ **e** $\frac{5}{8}-\frac{2}{5}$ **f** $\frac{5}{6}-\frac{3}{4}$

g $12\frac{3}{7}-2\frac{9}{10}$ **h** $6\frac{6}{7}-3\frac{4}{5}$ **i** $7\frac{2}{7}\times 6$

j $1\frac{2}{3}\times 3\frac{1}{4}$ **k** $2\frac{7}{9}\times 3\frac{4}{5}$ **l** $8\frac{1}{2}\times 12\frac{3}{8}$

m $2\frac{4}{5}\div 4$ **n** $4\frac{4}{9}\div 2$ **o** $2\frac{1}{3}\div 3\frac{3}{4}$

p $12\frac{1}{9}\div 3\frac{6}{7}$ **q** $\frac{1}{5}$ of 80 **r** $\frac{3}{4}$ of 420

s $\frac{2}{7}$ of 112 **t** $3\frac{2}{5}$ of 600 **u** 25% of 94

v 8% of 96 **w** 12.5% of 1200 **x** 0.8% of 250

7 Express:

a 15 as a percentage of 60 **b** 36 as a percentage of 54

c 3.2 as a percentage of 0.8 **d** 67 as a percentage of 67

e 0.38 as a percentage of 0.94 **f** $\frac{1}{3}$ as a percentage of $\frac{1}{7}$.

8 Increase:

a 100 by 70% **b** 25 by 12% **c** 11 by 3.5%

d 88 by $\frac{1}{4}$% **e** \$30 by 20% **f** 3 kg by 1%

9 Decrease:

a 50 by 25% **b** 800 by 3%

d 88.8 by 20% **d** 35 by 3.2%

e \$50 by 10% **f** 90 seconds by 33.3%

10 Estimate the value of each of the following.

a $\sqrt{6.1+2.9}$ **b** 14.6×2.7 **c** $46.2\div 25.3$

d 23.4^2 **e** 125×384 **f** $\frac{36.5+28.2}{29.9+4.8}$

g $\sqrt{49.1\times 24.8}$ **h** $\frac{\sqrt{99.6}}{\sqrt{143}}$

Core revision exercises: Algebra

Worksheet 6: Equations and transforming formulae

1 Factorise:

a $6x-6y$ **b** $12x-9y+3c$ **c** $6ax+3bx-9cx$

d x^2y+xy^2 **e** $3x^2+15xy$ **f** $x^4y^3+7x^2y-3x^3y^3$

g $2ax+4ay-3bx-6by$ **h** $12x^3-8x^2+2x$ **i** $\frac{1}{4}a+\frac{3}{4}b$

j $\frac{7}{8}x^3+\frac{3}{4}x^2$ **k** $3(x-4)+5(x-4)$ **l** $3(x-1)^2-18(x-1)^3$

2 Expand and simplify as far as possible.

a $-3(x-2y)$ **b** $-2x(3x-2)$ c $-4x(x-3)$

d $-5x(2x-2y)$ **e** $-2x(3x-7y)$ **f** $-8(y-7)$

g $-2x(x^2-5y)$ **h** $-2.4(3x-5)$ **i** $3(x-2)-(x-3)$

j $-2x(x+2)-x(3-x)$ **k** $-x(3-y)-2(y+2)$ **l** $-4(5x-2y)+3(4y-2x)$

3 Solve the following equations

a $x-9=18$ **b** $100x=4$ **c** $2x+7=35$

d $3x+9=17$ **e** $2x-4=-8$ **f** $4x=-2(x+2)$

g $3(5x-2)=-10$ **h** $-2(3-2x)=12$ **i** $2x+3=x-4$

j $3x-5=2x-1$ **k** $6-2x=4x-6$ **l** $4x+18=7x-3$

m $\frac{x}{2}-3=9$ **n** $\frac{2x}{5}-4=12$ **o** $x+\frac{4}{5}=6$

p $\frac{x}{3}-3=\frac{2x}{5}$ **q** $\frac{x}{3}-2=5$ **r** $\frac{3x}{2}+\frac{x}{3}=22$

s $\frac{x}{3}-8=6+\frac{x}{4}$ **t** $\frac{5x}{8}-2=\frac{x}{2}$

4 Solve the following equations.

a $4(x-2)=3(x-3)$ **b** $4(x+3)=7(4-2x)$

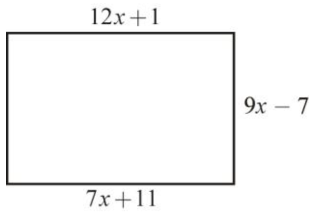
c $2-3(x-3)=3(x+4)$ **d** $3(1-x)=3(2x-3)+17$

e $3x-(2+x)=4-2(3-x)$ **f** $4-5(x-6)=3(3-x)-1$

g $x-1=3(3x-2)-(3x+4)$ **h** $3(x-2)-3(x-1)=4-6x$

i $4(x-2)-3(3+x)=4-x$

5 The lengths, in centimetres, of three sides of a rectangle are shown in the diagram.



Find the value of x and calculate the area of the rectangle.

6 Change the subject of each formula to the letter given in square brackets.

a $v = u + at$ $[t]$

b $s = x + y + z$ $[y]$

c $fh = g$ $[f]$

d $ab + c = d$ $[a]$

e $\frac{x}{y} = z$ $[x]$

f $y = x - 3$ $[x]$

g $S = \frac{D}{T}$ $[D]$

h $\frac{x}{y} = \frac{m}{n}$ $[m]$

i $a - \frac{b}{c} = x$ $[b]$

j $\sqrt{x} = y$ $[x]$

k $\sqrt{xy} = z$ $[y]$

l $x\sqrt{y} = a$ $[y]$

m $\sqrt{x} + y = m$ $[x]$

n $\sqrt{x} - y = b$ $[y]$

o $y\sqrt{x} = m$ $[x]$

p $\frac{x}{y} = \frac{p}{q}$ $[y]$

Core revision exercises: Shape, space and measures

Worksheet 7: Perimeter, area and volume

1 Calculate the circumference of a circle with diameter:

a 21 m　**b** 91 mm　**c** 3.4 cm

d 4.08 cm　**e** 1.8 m　**f** 2.5 cm

g 14.24 m　**h** 88 cm　**i** 10 m

j $16\frac{1}{2}$ cm　**k** x cm　**l** $(x+3)$ cm

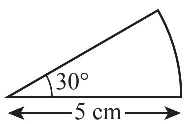
2 What is the radius of a circle of circumference:

a 14 mm　**b** 81 cm　**c** 206 mm

d 31.5 cm　**e** 220 cm　**f** x cm

3 What is the length of the arc in each of these sectors? Give your answer to 2 decimal places.

a



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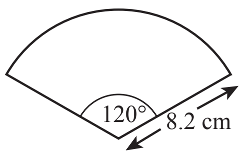
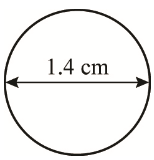


4 Find the area of the following shapes:

a



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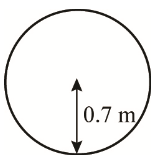

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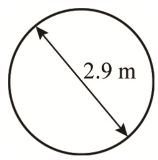

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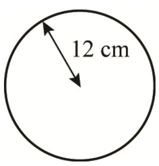

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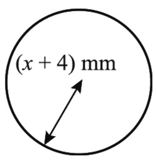
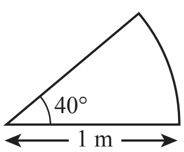


5 Calculate the area of each sector. Give your answers to 3 decimal places.

a



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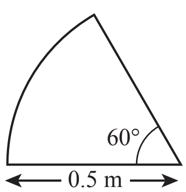
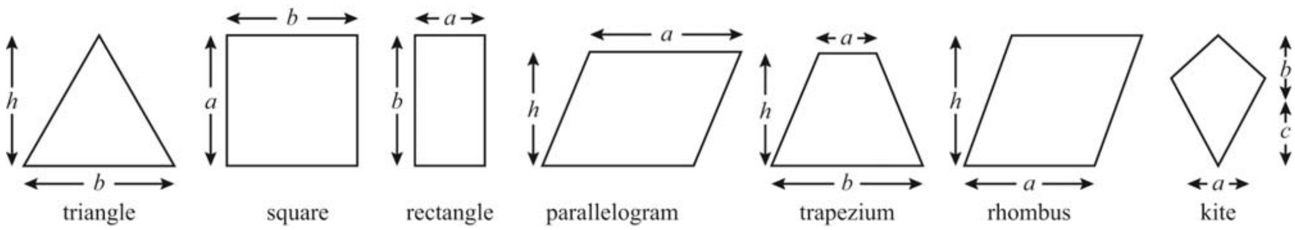


6 This diagram shows a silver pendant with a gold frame around it. The inner silver pendant has a diameter of 20 mm and the gold frame around it is 2 mm wide.

- **a** Giving your answer as a multiple of π, calculate the circumference of:
  - **i** the inner silver pendant
  - **ii** the outer gold ring.
- **b** What is the area of the silver part of the pendant? Give your answer in terms of π.

7 Write the formula for finding the area of each shape.



8 Calculate the area of each of the shapes listed below. Give your answers correct to two decimal places.

- **a** a square of side 12.6 cm
- **b** a rectangle with sides of 8.5 m and 12.2 m
- **c** a trapezium of height 12 cm and parallel sides of 8.5 cm and 11.8 cm.

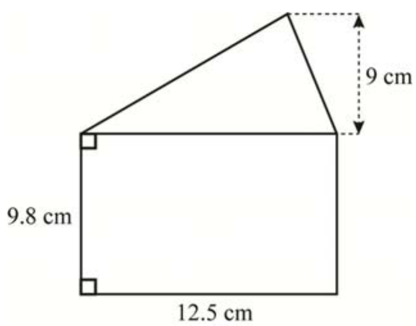
9 A triangular sail has a height of 5.65 m and an area of 68.93 m^2. What is the length of its base?

10 A square has an area of 529 cm^2. Calculate:

- **a** the length of a side
- **b** the perimeter of the square.

11 Find the area of each shape:

a



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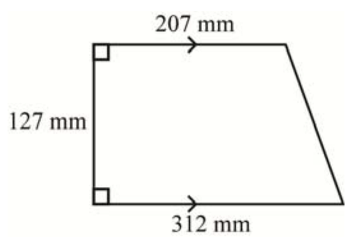

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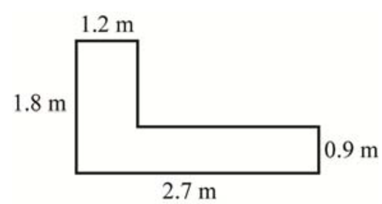

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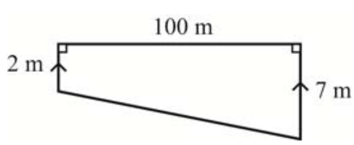

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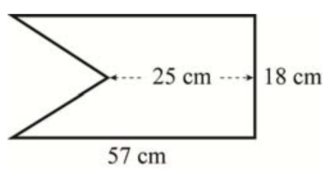
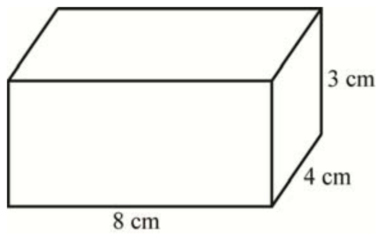


12 Calculate the volume of each of these solids.

a



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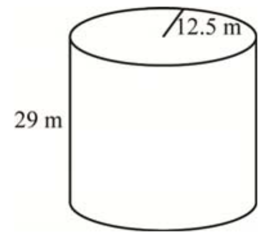

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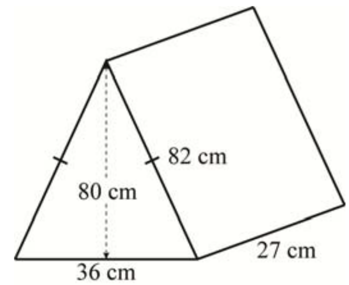

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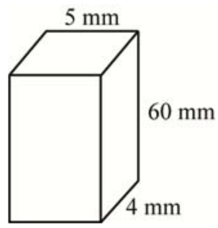

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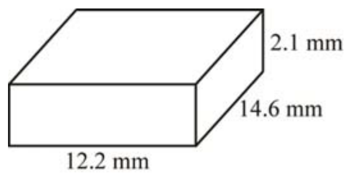

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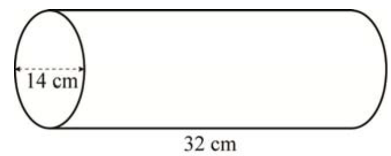

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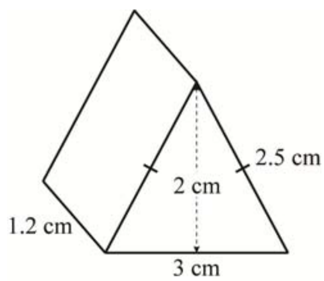

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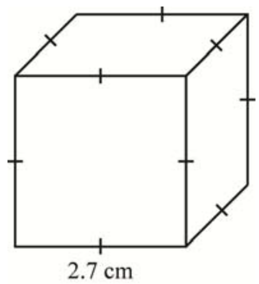
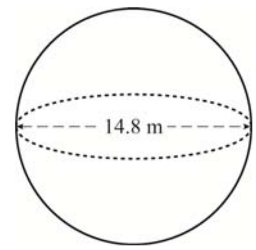


i



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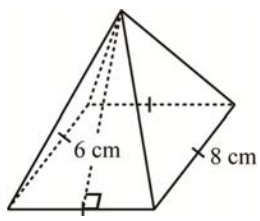

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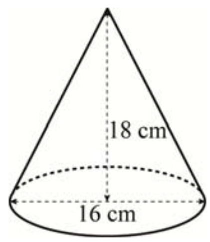

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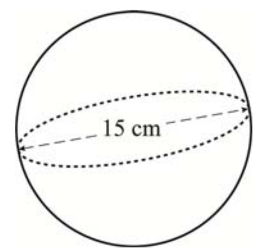


13 Draw the nets of shape **a** and shape **c** from question **12**.

14 Find the surface area of each solid in question **12**.

15 The radius of a cylinder is 90 cm. Its height is equal to $2r$ (where r is its radius). What is its surface area? Give your answer correct to 3sf.

16 A rectangular box measures 280 mm × 140 mm × 150 mm. How many smaller cuboids measuring 10 mm × 10 mm × 20 mm could be packed into the box?

Core revision exercises: Data handling

Worksheet 8: Introduction to probability

1 A box contains 3 red counters, 4 green counters, 2 blue counters and a black counter.

If you pick a counter without looking, what is the probability that you will pick:

a a red counter

b a green counter

c a black counter

d a blue counter

e a yellow counter

2 Nick shuffles an ordinary pack of 52 playing cards.

If he draw a single card at random, find the probability that is:

a red

b black

c spades

d not spades

e a king

f not an ace

g an even-numbered card

h a red even-numbered card

i not an even-numbered card.

3 A pentagonal spinner is divided into five equal segments. These are labelled A to E.

a Draw and label an accurate version of the spinner.

b Calculate the probability that it will land on:

i E　　**ii** G

iii Not A　　**iv** a vowel

v not a vowel.

4 A letter is chosen randomly from the word MATHEMATICS. What is the probability that it is:

a C　　**b** A

c not T?　　**d** a vowel?

e N?

5 A letter is randomly chosen from the song title 'You are my sunshine'.
What is the probability of the letter being:

a a capital letter?

b not a capital letter?

c an a?

d a consonant?

e not a consonant?

6 When you toss a fair six-sided dice, what is the probability of getting:

a an odd number

b an even number

c a prime number

d a multiple of 5

e not a multiple of 2

f not a 6

g a 7

h a factor of 36?

7 In a car park there are 35 red, 42 white, 12 black and 29 silver cars. 24 parking spaces are empty. What is the probability that a parking space chosen at random will contain:

a a red car

b a silver car

c not a black car

d no car at all?

8 Draw unbiased spinners that will land on blue, given the following information:

a $P(\text{blue}) = \frac{1}{6}$, $P(\text{red}) = \frac{5}{6}$

b $P(\text{blue}) = \frac{1}{3}$, $P(\text{white}) = \frac{1}{3}$, $P(\text{black}) = \frac{1}{3}$

c $P(\text{not blue}) = \frac{1}{8}$

d $P(\text{black}) = \frac{4}{5}$, $P(\text{blue}) = P(\text{not black})$

9 An unbiased black dice and an unbiased white dice are thrown together.

a Draw a probability space diagram to show all possible outcomes of this event.

b Find the probability that:

i one dice shows 2 and the other shows 3

ii one dice shows 6

iii both dice show 2.

Core revision exercises: Number

Worksheet 9: Sequences and sets

1 Write down:

a the 9th odd number **b** the 14th even number

c the 10th multiple of 8 **d** the 5th prime number

e the first 5 square numbers **f** the first 5 multiples of 9.

2 Write down the next three terms in each of the following sequences.

a $-2, -4, -6, -8, \ldots$ **b** $12, 9, 6, 3, \ldots$

c $1, 4, 9, 16, \ldots$ **d** $0, 6, 12, 18, \ldots$

e $12, 14, 18, 116, \ldots$ **f** $0.5, 1, 1.5, 2, 2.5, \ldots$

3 Find the nth term in each of the following sequences:

a $6, 8, 10, 12, \ldots$ **b** $2, 7, 12, 17, 22, \ldots$

c $2, 4, 6, 8, 10, \ldots$ **d** $19, 17, 15, 13, \ldots$

e $0.25, 1, 1.75, 2.5, 3.25, \ldots$ **f** $9, 7.8, 6.6, 5.4, \ldots$

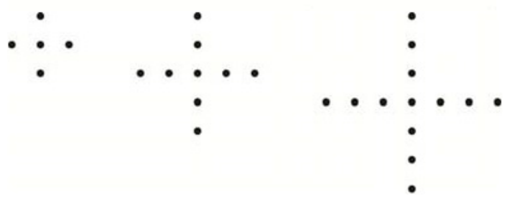
4 Here are some patterns made from dots.

a

b

c

d



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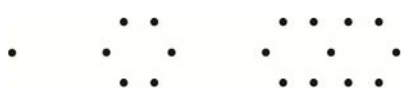


f

For each:

i Draw up a sequence table for the first six elements in each pattern, taking care to use the correct letter for the pattern number and the correct letter for the number of dots.

ii Find a formula for the number of dots used in terms of the pattern number.

iii Use your formula to find the number of dots in the 250th pattern.

5 The formula for the nth term of a sequence is $3n + 2$.

a Write down the first five terms in the sequence.

b Find:

i the 25th term

ii The 200th term

iii The 5000th term.

6 State whether or not the following numbers are rational or irrational. Give reasons for your answers.

a $\frac{3}{7}$

b $\sqrt{8}$

c $\sqrt{9}$

d $3-\sqrt{3}$

e $\sqrt{(3-3)}$

f $0.\dot{6}\dot{3}$

g $1.\dot{8}$

h $\frac{\sqrt{8}}{\sqrt{2}}$

i π^2

j $\left(\sqrt{17}\right)^2$

k $\left(\sqrt{17}\right)^3$

l $\frac{1}{\pi}$

m $\sqrt{\frac{1}{9}}$

n $\sqrt{\frac{2}{3}}$

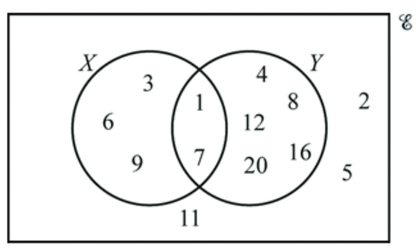
o $\sqrt{0.09}+\sqrt{2.25}+\sqrt{0.49}$

7 P = {9, 10, 11, 12, 13, 14, 15, 16}

a Describe this set in words.

b What is n(P)?

c List R $\cap$ S.



8 List the members of the following sets:

a X

b Y

c X $\cap$ Y

d X $\cup$ Y

9 For the sequence $u_n = \frac{1}{2}n(n+1)$:

a list the first five terms

b identify the special sequence of numbers

Core revision exercises: Algebra

Worksheet 10: Straight lines and quadratic equations

1 Complete each table of values for the given values of x.

a $y = x + 3$

x	-1	0	1	2	3
y					

b $y = 3$

x	-1	0	1	2	3
y					

c $y = \frac{1}{2}x - 1$

x	-1	0	1	2	3
y					

d $y = 2x - 4$

x	-1	0	1	2	3
y					

e $y = -\frac{1}{2}x$

x	-1	0	1	2	3
y					

f $y = -2x + 1$

x	-1	0	1	2	3
y					

g $y = x - 1$

x	-1	0	1	2	3
y					

h $2x - y = 4$

x	-1	0	1	2	3
y					

i $x = 7$

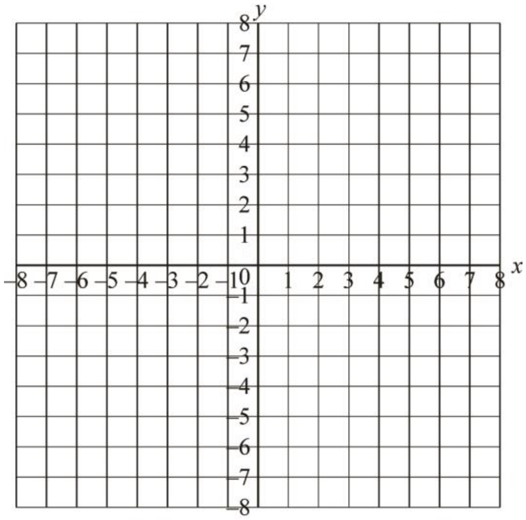
x					
y	-1	-1	1	2	3

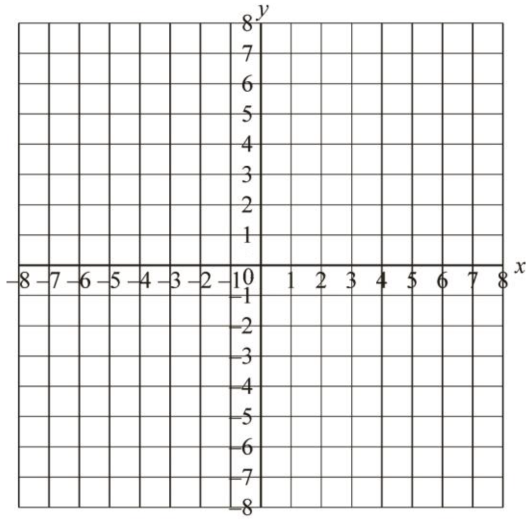
j $x + y = -1$

x	-1	0	1	2	3
y					

2 Here are two sets of axes.

For the tables produced in question 1, draw and label graphs (a) to (e) on one set of axes and draw and label graphs (f) to (j) on the other set of axes.

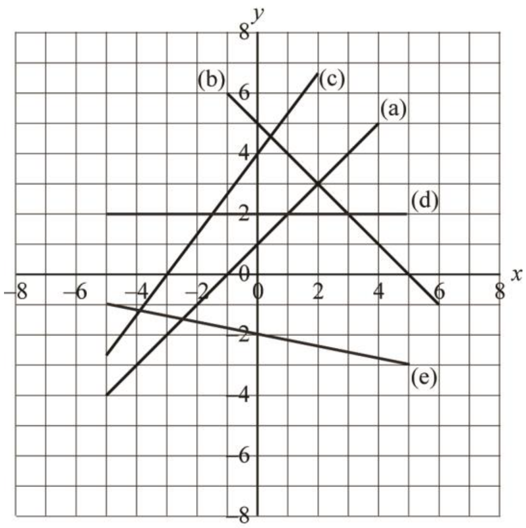




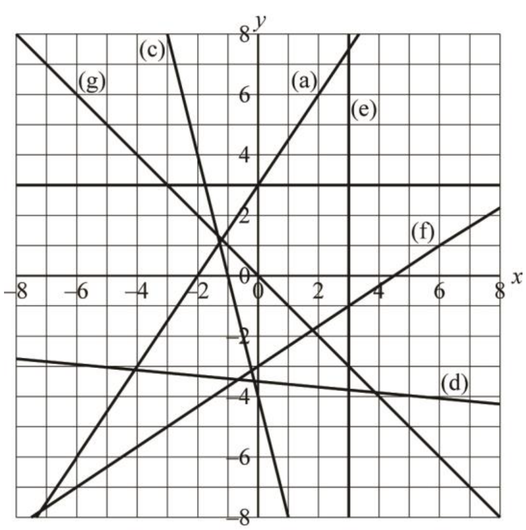
3 For each line shown:

i find the gradient of the line

ii find the equation of the line.



4 Find the equation of each line.



5 Find the equation of a line that is:

- **a** parallel to the line with equation $y = 4x + 1$, but passes through the point (3, 16)
- **b** parallel to the line with equation $y = -3x + 5$, but passes through the point (7, – 8)
- **c** parallel to the line with equation $y = 0.5x + 0.3$, but passes through the point (3, 2.4)
- **d** parallel to the line with equation $3x + 4y = 12$, but passes through the point (2, –1)
- **e** parallel to the line with equation $5x - 2y = 18$, but passes through the point (– 3, –4).

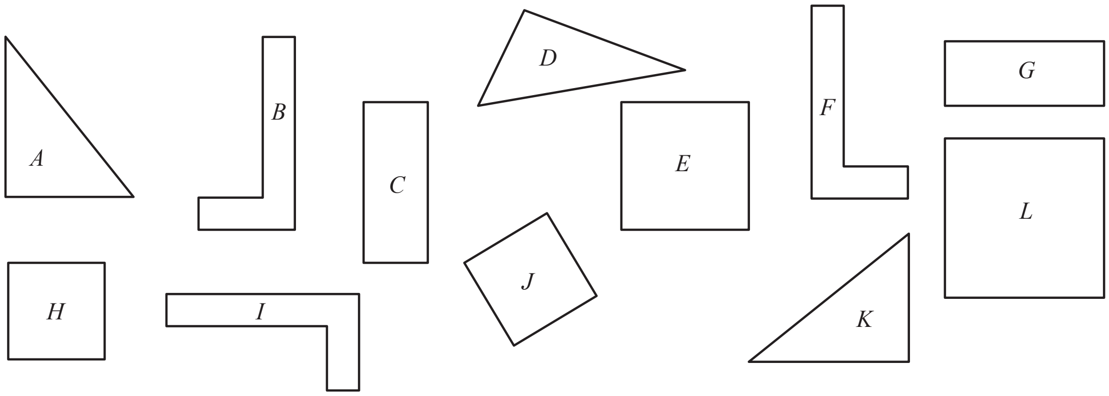
6 Expand and simplify each of the following.

a	$(x + 1)(x + 11)$	**b**	$(x - 7)(x + 12)$	**c**	$(y + 3)(y - 4)$
d	$(x - 4)(x + 4)$	**e**	$(p - 8)(p - 9)$	**f**	$(x - 3)^2$
g	$(h + 5)(h - 15)$	**h**	$(p - 3)(p - 10)$	**i**	$(2x + 3)(2x - 4)$

Core revision exercises: Shape, space and measures

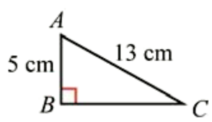
Worksheet 11: Pythagoras' theorem and similar shapes

1 Which of the following figures are congruent?



2 Find the length of the unmarked side in each of the following triangles.

a



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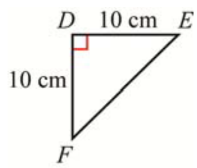

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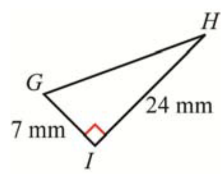

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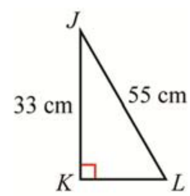

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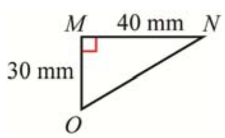

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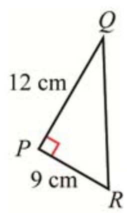

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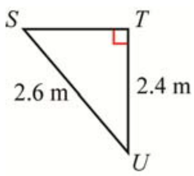

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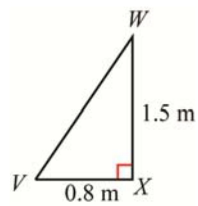
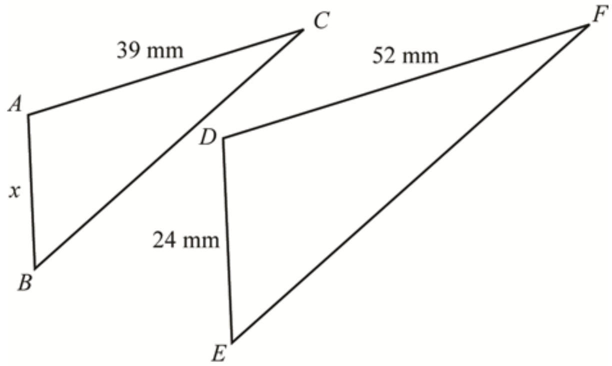


3 A rectangular field is 120 m long and 88 m wide.
What is the longest distance it is possible to walk in the same direction in one continuous straight line?
Give your answer correct to the closest metre.

4 Josh and Sarah walk from the same point. Josh walks due west and Sarah walks due north.
After 1 hour, Josh is 4.2 km from the starting point and Sarah is 5.6 km from Josh in a straight line.
How far is she from the starting point?

5 The following pairs of triangles are similar.
Find the length of each side marked with a letter.

a



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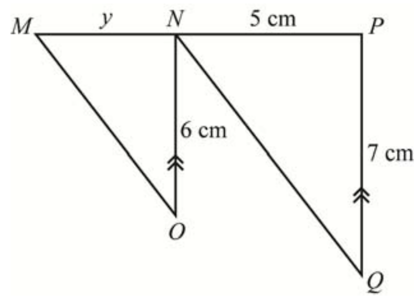

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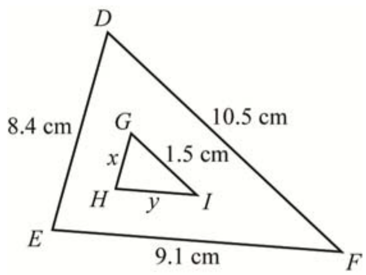

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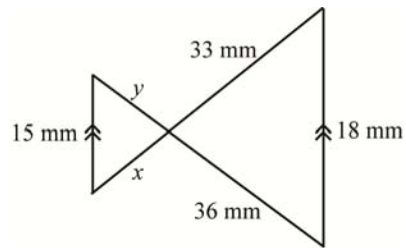
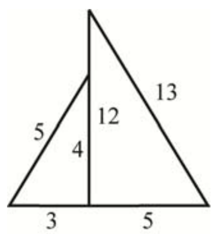


6 Determine whether each pair of triangles is similar or not.
Show your working.

a



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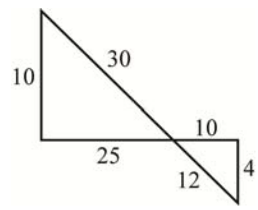

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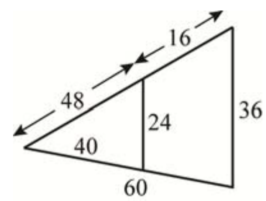

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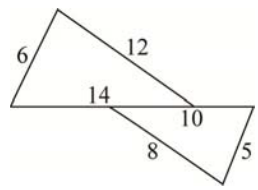
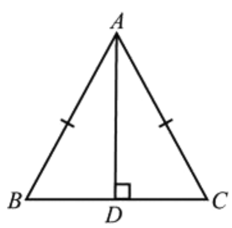


7 Which of the following pairs of triangles are congruent?

a



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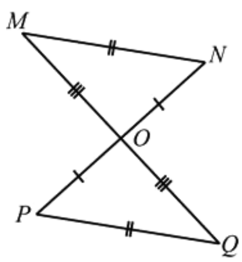

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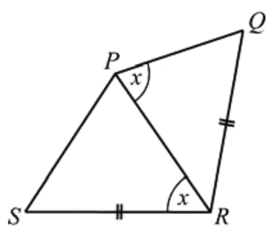

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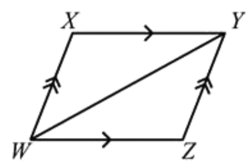

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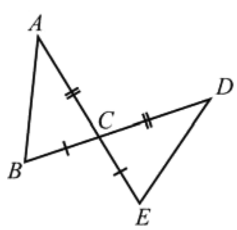

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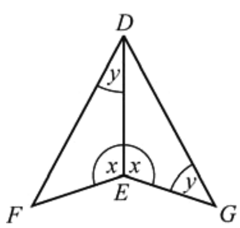

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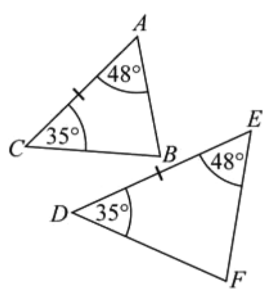

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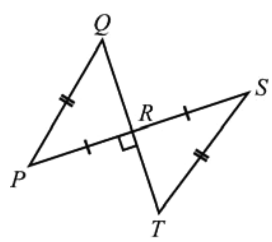


Core revision exercises: Data handling

Worksheet 12: Averages and measures of spread

1 Find the mean, median, mode and range of the following sets of data.

a 1, 2, 3, 5, 5, 6, 8, 3, 4, 3, 7, 8

b 1, 2, 2, 2, 3, 4, 5, 6, 6, 6, 7, 8, 8, 9, 10

c 1, 3, 12, 14, 13, 22, 10, 11, 12, 4, 5, 7

d 2, 3, 4, 3, 5, 5, 6, 7, 8, 9, 6, 6, 5, 3

e 13, 7, 8, 8, 2, 9, 11, 7, 8, 4, 5

f 45, 48, 60, 42, 53, 47, 51, 54, 49, 48, 47, 53, 48, 44, 46

g 11.0, 11.2, 11.4, 11.0, 11.8, 11.8, 11.4, 12.0, 11.8, 11.6

h 64, 70, 70, 72, 76, 76, 77, 77, 77, 78, 78, 80

2 The stem-and-leaf diagram shows how many minutes Sven spends on social media each day for four weeks.
Calculate the mode, median and range of the data.

Stem	Leaf
0	5 7 8 9 9
1	1 1 1 3 5 7 9
2	1 2 5 7
3	1 1 8 9
4	0 3 4 9
5	1 1 4 8

Key 0 | 5 = 5 minutes
1 | 1 = 11 minutes

3 Complete these frequency tables and calculate the mean, median and mode for each set of data.

a

Outcome	Frequency	fx
1	**2**	
2	**4**	
3	**6**	
4	**3**	
5	**2**	
6	**2**	

b

Outcome	Frequency	fx
123	**8**	
124	**4**	
125	**5**	
126	**6**	
127	**7**	
128	**8**	
129	**9**	

c

Outcome	Frequency	fx
7	**12**	
8	**9**	
9	**16**	
10	**13**	
11	**20**	
12	**20**	

4 The mean of seven scores is 18.

a What is the total of the scores?

b The lowest score is 12 and the range of scores is 11. What is the highest score?

5 Two students get the following results for six mathematics tests (out of 100):

Anna: 60, 90, 100, 90, 90, 100

Zane: 60, 70, 60, 70, 70, 100

a What is the range of scores for each student?

b Does this mean they both had equally good results?

c Which statistic would be a better measure of their acheivement? Why?

6 The shoe sizes of 50 students are given below:

10	8	6	6	5	4	6	7	8	6
9	6	7	8	9	10	4	4	5	6
7	7	6	4	8	9	10	6	7	6
5	5	5	6	4	7	8	9	6	7
5	4	4	6	6	6	7	8	9	9

a Draw up a frequency table to show this data.

b What is the modal shoe size?

c What is the mean shoe size?

d What it the median shoe size?

e Which value is most useful for understanding this data? Why?

7 In April 2017, the following statistic appeared in a newspaper:

'The median age in a sample of developing countries was 19 in 1990. It is now 24 and will be 30 by 2030.'

a What does the term 'median age' mean?

b What does this statistic tell you about the population in these countries?

Core revision exercises: Measurement

Worksheet 13: Understanding measurement

1 Convert the following length measurements to the units given:

a 4.5 km = ___ m **b** 97 cm = ___ mm

c 9.7 m = ___ cm **d** 2324.65 m = ___ km

e 0.08 m = ___ mm **f** 5.9 cm = ___ m

2 Convert the following measurements to the units given:

a 5.8 kg = ___ g **b** 26.67 kg = ___ g

c 0.9 kg = ___ g **d** 77 g = ___ kg

e 12.5 g = ___ kg **f** 3 000 000 g = ___ tonne

3 Underline the greater length in each of these pairs of lengths. Then calculate the difference between the two lengths. Give your answer in the most appropriate units.

a 9 km, 8900 m **b** 690 m, 69 015 cm

c 28.3 cm, 285 mm **d** 401 cm, 4.99 m

e 685 m, 0.7 km **f** 5.5 km, 545 000 cm

4 Convert the following area measurements to the units given:

a 7 cm^2 = ___ mm^2 **b** 12 cm^2 = ___ mm^2

c 64.2 cm^2 = ___ mm^2 **d** 0.07 km^2 = ___ m^2

e 9543 m^2 = ___ km^2 **f** 4.423 m^2 to mm^2

5 Convert the following volume measurements to the units given:

a 56 cm^3 = ___ mm^3 **b** 11 cm^3 = ___ mm^3

c 34.4 cm^3 = ___ mm^3 **d** 485 cm^3 = ___ mm^3

e 149 mm^3 = ___ cm^3 **f** 19 600 mm^3 = ___ m^3

6 **a** Complete this time-card to show how long a shop assistant worked each day for a week.

Day	Time in	Time out	Lunch	Hours worked
Monday	8.15 a.m.	5.25 p.m.	45 mins	
Tuesday	8.17 a.m.	5.30 p.m.	30 mins	
Wednesday	8.23 a.m.	5.50 p.m.	45 mins	
Thursday	8.22 a.m.	6.00 p.m.	60 mins	
Friday	7.58 a.m.	7.00 p.m.	45 mins	

b Calculate the total hours worked for the week.

c If this person earned \$4.95 per hour, calculate how much he would have earned this week.

d The employer has to deduct 12% tax from the shop assistant's earnings. How much would he be paid after deductions?

7 Each of the numbers below has been rounded to the degree of accuracy shown in the brackets. Find the upper and lower bounds in each case.

a 34 (nearest unit) **b** 12 878 (nearest unit)

c 600 (1sf) **d** 15.34 (2dp)

e 12.69 (2dp) **f** 4.5 (to nearest 0.5)

g 670 (nearest 10) **h** 3.142 (3sf)

8 A child is weighed at a clinic and her mass is recorded to the nearest half kilogram as 12.5 kg. What is the greatest and least possible mass of this child?

9 These exchange rates are displayed at a hotel reception desk in Mumbai.

Currency	Rate (Indian Rupees)
US \$	44.62
UK £	72.18
Euro (€)	64.24
Australian \$	46.58
UAE Dirham (Dhs)	12.09
Saudi Riyal (SR)	11.84

a Use the table to convert these amounts of money to Indian Rupees.

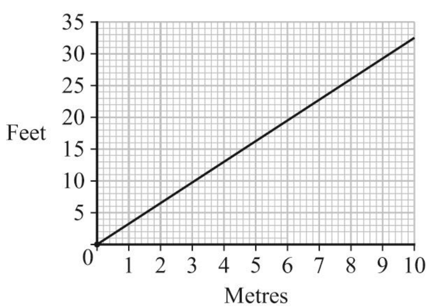
i US\$100 **ii** Aus\$75 **iii** £145

iv €600 **v** Dhs450 **vi** SR1265

b Sheik Abulla bin Mohammed stayed at the hotel. His bill in Indian Rupees was Rs45 600. What is this in Saudi Riyals?

c Mrs Piccolo, an Australian, booked a room for five nights at a rate of Rs14 000 per night. What would her accomodation bill be for the five nights in Australian dollars?

10 This graph shows the relationship between length in metres (m) and length in feet (ft).



a What does one small square on the horizontal axis represent?

b What does one small square on the vertical axis represent?

c Change 3 metres to feet.

d The ceiling in a hall is 15 feet above the floor. How many metres is this?

e Sandra is approximately six foot tall. How tall is this in metres?

Core revision exercises: Algebra

Worksheet 14: Further solving of equations and inequalities

1 Solve the following pairs of simultaneous equations.

a $2x + y = 17$
$3x - y = 13$

b $x + 2y = 12$
$x + 6y = 30$

c $2x - y = 16$
$4x - y = 28$

d $14x - 8y = 2$
$-6x + 8y = -10$

e $y = x + 3$
$y = 3x - 7$

f $2x + 5y = 19$
$4x + 5y = 23$

g $4x + 4y = 24$
$4x + y = 15$

h $3x + 3y = -3$
$3x - y = 7$

i $y + 2x = 8$
$4y - 3x = -12$

j $x - 3y = 0$
$2x - y = 20$

k $3y - x = 11$
$2y - 3x = 5$

l $5x - y = 31$
$6x + 3y = 21$

m $x = 12 - y$
$4y - 3x = -12$

n $2x + y = 3$
$6x + 2y = 0$

o $2x + y = 11$
$3x = 9 + y$

p $4x = 2y + 6$
$4x = 14 - 2y$

q $2x + 2y = -4$
$3x - 5y = -38$

r $14x + 6y = 8$
$10x + 4y = 6$

s $x - 4y = -2$
$3x + 5y = 45$

t $7x - 4y - 6 = 0$
$3x + y = 4$

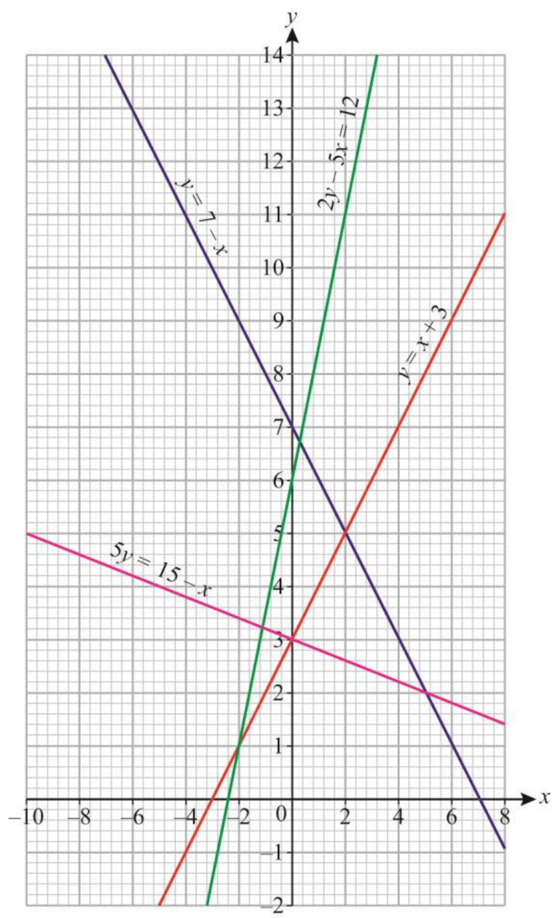
2 Use the graph below to find the solution to the following pairs of simultaneous equations.

a $y = x + 3$
$y = 7 - x$

b $y = 7 - x$
$5y = 15 - x$

c $5y = 15 - x$
$y = x + 3$

d $y = x + 3$
$2y - 5x = 12$



3 Draw accurate graphs and use them to solve each pair of simultaneous equations given below. Check your answers by substituting the coordinates in the original.

a $y = x + 3$
$x + 4y = 22$

b $y = x - 2$
$x + 4y = 12$

c $y = x + 1$
$x + y = 2$

d $y = \frac{1}{2}x - 1$
$x + 2y = 0$

Core revision exercises: Shape, space and measures

Worksheet 15: Scale drawings, bearings and trigonometry

1 Draw lines (in cm) to show how long each of the following lengths would be on a scale drawing with a scale of 1 : 250.

a 50 cm　　**b** 500 cm

c 850 cm　　**d** 3.5 m

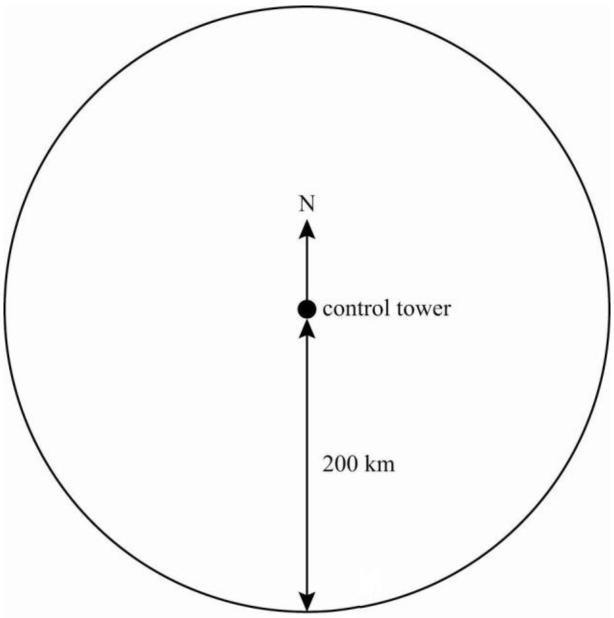
e 9000 mm　　**f** 0.0045 km

2 The bearings of five aeroplanes, all 200 km from a control tower, are given below. Use these to accurately indicate the position of each plane on the diagram.

i 065°　　**ii** 093°

iii 172°　　**iv** 268°

v 308°

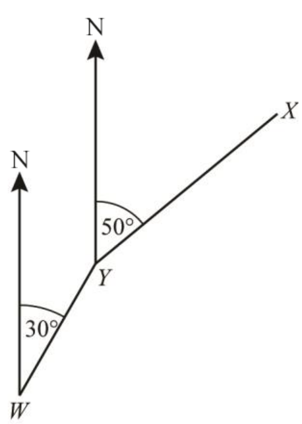


NOT TO SCALE

3 If the bearing of X from Y is 050° and the bearing of Y from W is 030°, calculate the size of:

a angle WYN

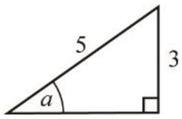
b angle WYX



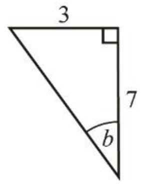
NOT TO SCALE

4 Find the size of the marked angle. Give your answers correct to one decimal place.

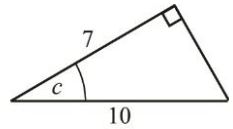
a



b



c



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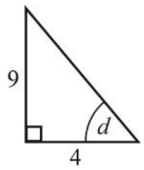

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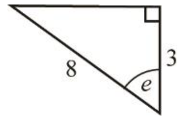

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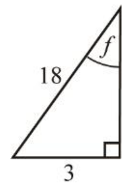
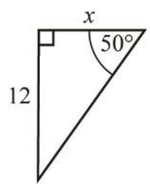
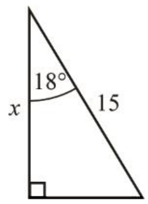


5 Find the length of the side marked x in the following triangles.

a



b



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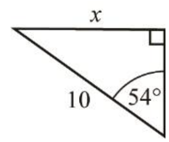

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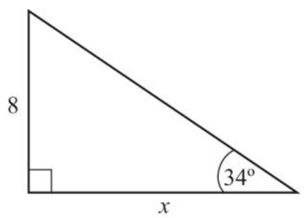

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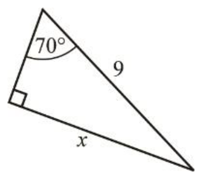

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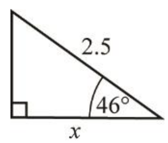


6 A hiking path slopes upwards at an angle of 18°.

If Cedric walks 600 m up the path, how high is he above his starting point?

7 A 9 m high vertical cellphone mast on a level concrete slab is supported by two stay wires 10 m long. Each stay wire is attached to the top of the pole and to the slab.

Calculate:

a The angle between the stay wire and the slab.

b The distance from the bottom of the mast to the point where the stay wires are attached to the slab.

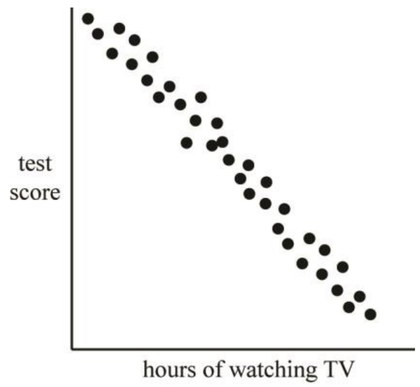
Core revision exercises: Data handling

Worksheet 16: Scatter diagrams and correlation

1 **a** Describe the correlation shown on the following scatter diagrams.

b Draw a line of best fit on graphs **a**, **b**, **d** and **e**.

i



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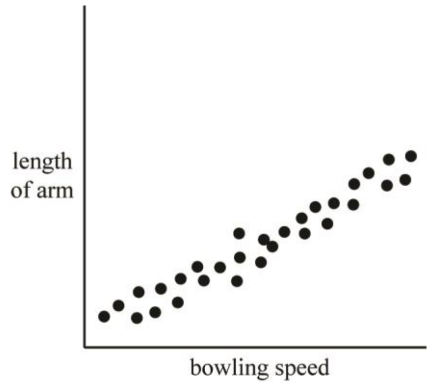

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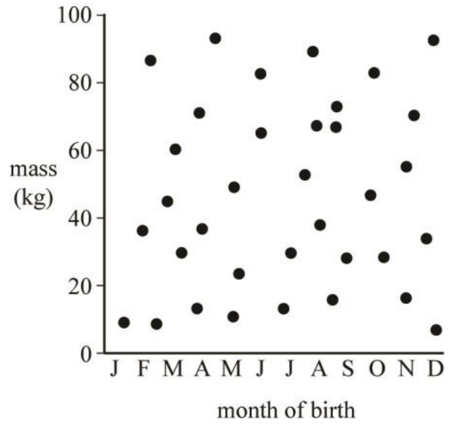

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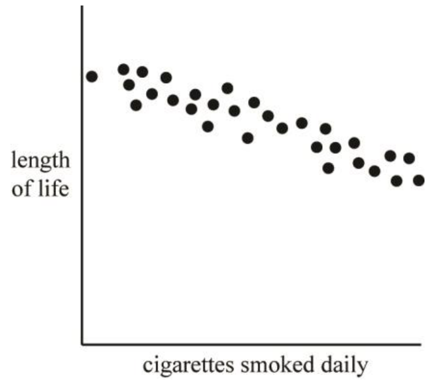

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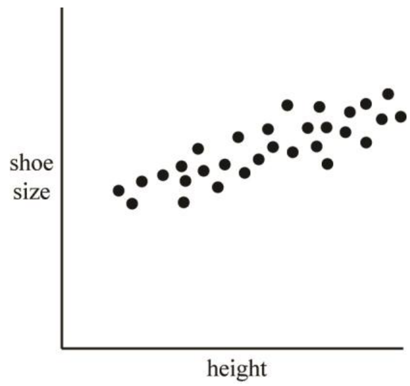


2 Mrs Almasri wants to know whether her students' results on a mid-year test are a good indication of how well they will do in the end-of-year test. The results from the test and the examination are given for a group of students.

Student	Mid-year mark	End-of-year test mark	Student	Mid-year mark	End-of-year test mark
Anna	78	73	Tina	92	86
Nick	57	51	Yemi	41	50
Sarah	30	39	Asma	75	64
Ahmed	74	80	Rita	84	77
Sanjita	74	74	Mike	55	58
Moeneeb	88	73	Karen	90	80
Kwezi	94	88	James	89	87
Pete	83	69	Priya	95	96
Idowu	70	63	Claudia	67	70
Sam	61	67	Noel	45	50
Emma	64	68	Wilma	70	64
Gibrine	49	54	Teshi	29	34

a Draw a scatter diagram with the end-of-year test results on the vertical axis.

b Comment on the strength of the correlation.

c Draw a line of best fit for this data.

d Estimate the end-of-year test results of a student who got 65 in the mid-year test.

e Comment on the likely accuracy of your estimate in part (d).

Core revision exercises: Number

Worksheet 17: Managing money

1 A sporting goods store bought 100 tennis racquets for \$60 000. They sold the racquets for \$800 each.

a Calculate the profit per tennis racquet.

b Calculate the percentage profit per racquet.

2 During the Shopping Festival in Dubai, a jeweller offers a 10% discount on all items.

Calculate the sale price of a necklace with a marked price of \$1200.

3 In the UK, value-added tax (VAT) of 20% is added to the price of goods before they are sold.

Work out the selling price of a computer that is marked at £1250 excluding VAT.

4 Selina wants to buy an entertainment system. The marked price is \$3000. There is a discount of 10% for cash payments. The goods can also bought on hire purchase with a deposit of \$800 and 18 monthly installments of \$160.

Calculate:

a the cash price of the entertainment system

b the total amount Selina would pay if she bought it on hire purchase

c the difference between the HP price and the cash price.

5 A woman who works a 38 hour week earns \$731.50.

What is her hourly rate of pay?

6 What is the annual salary of a person who earns \$2450 per month?

7 A plumber's assistant earns \$22.40 per hour for a 35 hour week. He is paid 1.5 times his hourly rate for each hour he works above 35 hours.

How much would he earn in a week if he worked:

a 36 hours **b** 40 hours **c** 30 hours **d** $42\frac{1}{2}$ hours.

8 Sandile earns a gross salary of \$1203.50 per month. His employer deducts 15% income tax, \$174.20 insurance and 1.5% for union dues.

What is Sandile's net salary?

9 Use the formula to calculate the compound interest earned if you invest:

a \$400 for 3 years at 5% **b** \$7000 at 12% for $4\frac{1}{2}$ years **c** \$4000 at 8.5% for 3 years.

10 A woman invests \$5000 in an investment scheme for 5 years and earns 8% pa simple interest.

a Calculate the total interest she will earn.

b How much would she need to invest to earn \$3600 interest in the same period (at the same rate)?

11 This table compares the simple and compound interest earned on a \$10 000 investment a rate of 9% pa.

Years	1	2	3	4	5	6	7	8
Simple interest	900	1800	2700	3600	4500	5400		
Compound interest	900	1881	2950.29	4115.82	5386.24	6771		

a Complete the last two columns of the table.

b What is the difference between the simple interest and compound interest earned after 5 years?

c Draw a bar graph to compare the value of the investment after 1, 5 and 8 years for both types of interest.

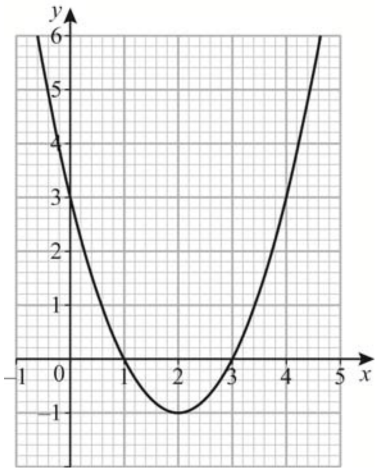
Core revision exercises: Algebra

Worksheet 18: Curved graphs

1 Construct a table of values, selecting values from $-6 \leqslant x \leqslant 6$, for each of the following equations. Draw the graphs on the same set of axes.

a $y = -x^2 + 3$ **b** $y = x^2 - 2x + 2$ **c** $y = x^2 + 8x + 16$

2 This is the graph of $y = x^2 - 4x + 3$. Use the graph to estimate the solution of:



a $x^2 - 4x + 3 = 0$

b $x^2 - 4x + 3 = 3$

c $x^2 - 4x = 1$

3 By drawing up a table of values, and drawing the graphs on the same set of axes, find the approximate solutions to the simultaneous equations $2y = x - 1$ and $y = x^2 - 4x + 3$.

4 For each of the following:

a complete the table of values for each equation

b use the points to plot each graph on a separate set of axes.

x	–5	–4	–3	–2	–1	1	2	3	4	5
$y = \frac{2}{x}$										

x	–5	–4	–3	–2	–1	1	2	3	4	5
$y = \frac{-1}{x}$										

x	–5	–4	–3	–2	–1	1	2	3	4	5
$xy = 1$										

x	–5	–4	–3	–2	–1	1	2	3	4	5
$xy = -2$										

5 Sketch the following graphs and label them.

a $y = 3x^2$

b $y = -x^2 + 4$

c $y = \frac{12}{x}$

Core revision exercises: Shape, space and measures

Worksheet 19: Symmetry

1 **a** How many lines of symmetry are there for each letter in this word in the font shown here?

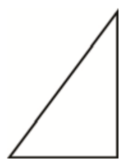
CAMBRIDGE

b Which letters in the word have rotational symmetry and what is the order of rotation?

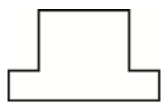
2 Draw a quadrilateral with two lines of symmetry and name it.

3 Each of these shapes is half of a symmetrical shape. Complete the shapes by drawing their other halves. Indicate the line of symmetry on your completed shapes.

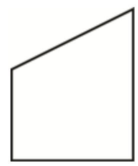
a



b

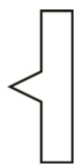


c



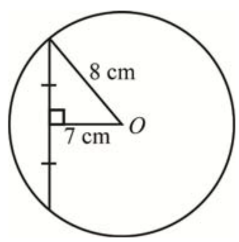
d

e

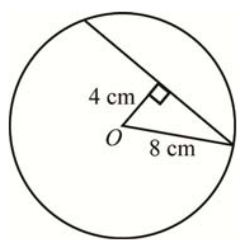


4 Find the length of chords shown in each diagram.

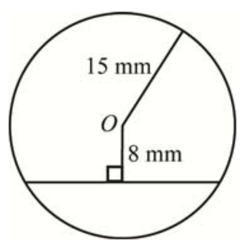
a



b

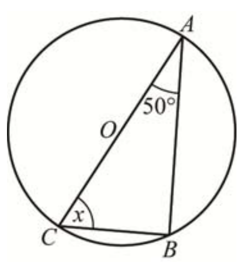


c



5 Find the size of angle x in each of the following diagrams. Show your working and give reasons for any statements you make.

a



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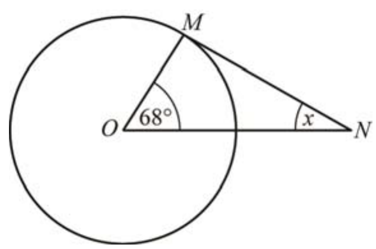

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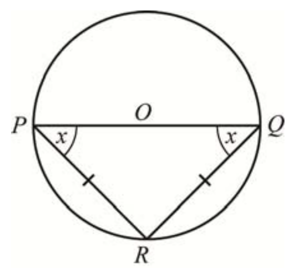
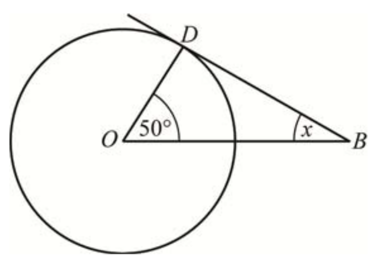
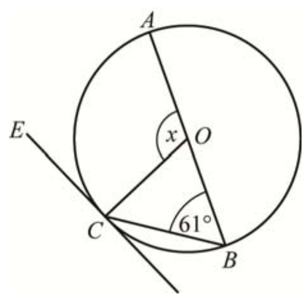


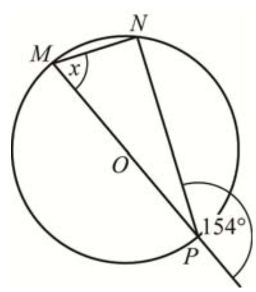
d



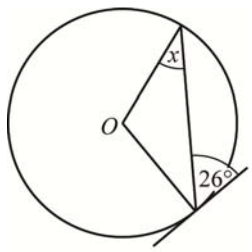
e



f



g



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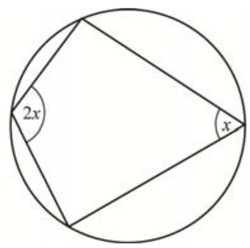
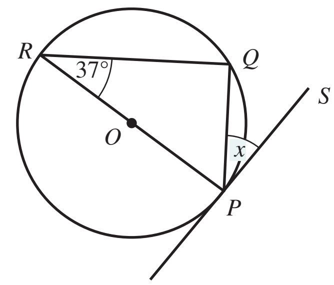


i



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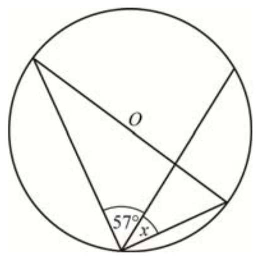
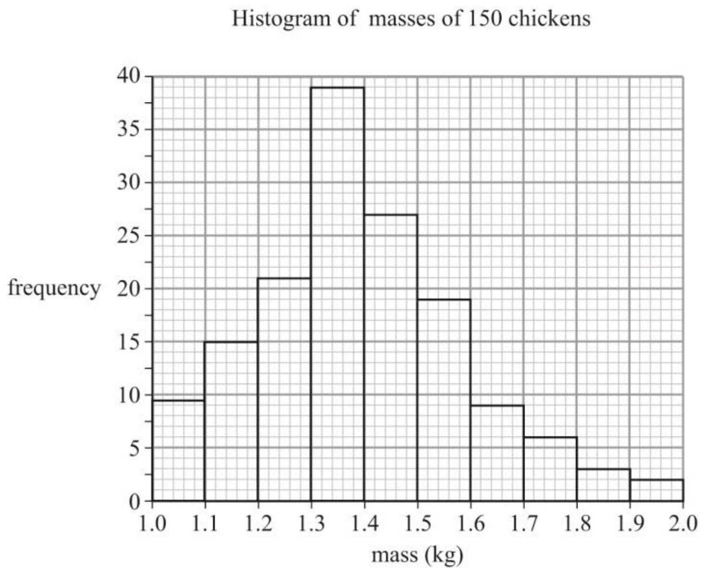


Core revision exercises: Data handling

Worksheet 20: Histograms and frequency distribution diagrams

1 This histogram shows the masses of 150 roast chickens sold at a deli counter in one week.



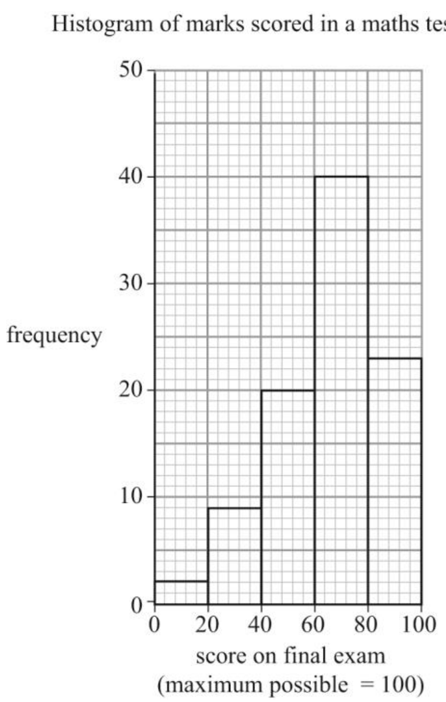
a What is the modal mass of the roast chickens?

b What range of masses was recorded?

c How many roast chickens had a mass of more than 1.8 kg?

d How many chickens had a mass between 1.3 and 1.6 kg?

2 This histogram shows the frequency of different marks in a mathematics test.



a What is the modal mark?

b A mark of 80 or higher is awarded an A. How many students achieved an A?

c Can you tell how many students scored less than 70% from this graph? Explain why or why not.

3 This table gives the mass, in kilograms, of 200 students. Draw a histogram to show this distribution.

Mass (kg)	$40 \leqslant m < 50$	$50 \leqslant m < 60$	$60 \leqslant m < 70$	$70 \leqslant m < 80$	$80 \leqslant m < 90$
Frequency	35	55	60	35	15

4 The ages of 33 young adults attending a youth camp are shown below.

12	15	17	18	16	17	18	19	20	18	11
16	13	14	15	17	19	18	17	19	18	12
13	15	15	16	17	19	20	18	15	15	23

a Organise the data using this frequency table.

Ages	Frequency
$10 \leqslant \text{age} < 15$	
$15 \leqslant \text{age} < 20$	
$20 \leqslant \text{age} < 25$	

b Draw a histogram to represent the ages of the students.

Core revision exercises: Number

Worksheet 21: Ratio, rate and proportion

1 Express each of the following as the rate given in brackets.

- **a** A journey of 423 km which took 2 hours and 15 minutes (km/h).
- **b** A cricketer scored 100 runs in 80 minutes (runs/min).
- **c** 6 metres of material cost $75.12 ($/m)

2 Express the following ratios in their simplest form.

- **a** $\frac{3}{8}:\frac{7}{8}$
- **b** 1:1.6
- **c** 2 : 2.25
- **d** 27 :18
- **e** $3\frac{1}{2}:7$
- **f** 45c to $2.20
- **g** 225 m to 0.425 km
- **h** 100 minutes to $1\frac{1}{2}$ hours

3 A person runs at a speed of 5 m/s. How far will he run in:

- **a** 10 seconds?
- **b** 1 minute?
- **c** $2\frac{1}{2}$ minutes?

4 Water runs out of a pipe at a rate of 0.15 litres per second. How long will it take to fill a 25 litre container?

5 Concrete is made by mixing stone, sand and cement in the ratio 3: 2 :1.
If 14 wheelbarrows of sand are used, how much stone and cement is needed?

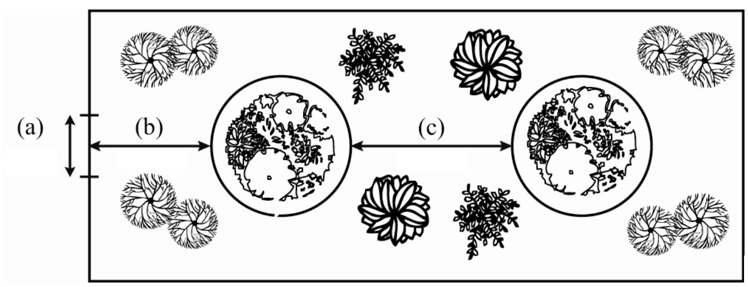
6 The distance between New Delhi and Mumbai is 1407 km. On a map of India, this distance measures 15.633 cm. What is the scale of the map?

7 In each of the following, calculate x.

- **a** $\frac{x}{10}=\frac{18}{15}$
- **b** $\frac{3}{35}=\frac{x}{28}$
- **c** $x : 24 = 49 : 56$
- **d** $0.8 : 20 = x : 4$

8 Square A and square B have sides of 125 mm and 6 cm respectively. Find the ratio of their areas.

9 This plan of a botanical garden is drawn at a scale of 1 : 750.



NOT TO SCALE

- **a** Measure the lengths indicated on the diagram in mm. Write each length on the diagram.
- **b** Use the scale to work out the actual length of each section indicated on the diagram.

10 How long will it take to travel 200 km at:

- **a** 75 km/h?
- **b** 80 km/h?
- **c** 45 km/h?
- **d** 120 km/h?
- **e** 4 m/min?

11 Emily has $600 to spend on books for the school library.

- **a** Complete this table of values to show how many books of different prices she can buy.

Price	**$1**	**$2**	**$5**	**$7.50**	**$10**	**$12**	**$15**	**$20**	**$25**	**$50**
No of books										

- **b** Draw a graph to show the relationship between the price of books and the number she can buy for $600.

Core revision exercises: Algebra

Worksheet 22: More equations, formulae and functions

1 Solve each problem by setting up an equation.

a When seven is added to four times a certain number, the result is 59.
What is the number?

b If six is subtracted from three times a certain number the result is 45.
What is the number?

c Four more than a number is divided by three and then multiplied by two to give a result of twelve.
What is the number?

d A number is doubled and then six is added. When this is divided by four, the result is eleven.
What is the number?

2 Represent each situation using an equation in terms of y.
Solve each equation to find the value of y.

a A number is multiplied by three, then five is added to get 19.

b When six is subtracted from five times a certain number, the result is 174.

c Three times the sum of a number and four gives 93.

d When eight is subtracted from half of a number, the result is 54.

3 A father is three times as old as his daughter.
If the father is 28 years older than his daughter, what are their ages?

4 Jess and Silvia have 174 marbles between them.
If Jess has five times as many marbles as Silvia, how many do they each have?

5 Soumik has $5 less than his friend Kofi.
If they have $237.50 altogether, how much does each person have?

6 Two competition winners are to share a prize of $825.
If one winner receives twice as much as the other, how much will they each receive?

7 A grandfather is five times as old as his grandson.
If the grandfather was 48 when his grandson was born how old is the grandson?

8 A rectangle of perimeter 120 cm is 7 cm longer than it is wide.
What is the length of each side?

9 Greenburg is located between Brownburg and Townburg. Greenburg is five times as far away from Townburg as it is from Brownburg.
If the distance between Brownburg and Townburg is 864 km, how far is it from Brownburg to Greenburg?

10 Amira is twice as old as her cousin Pam. Seven years ago, their combined age was 19.
What are their present ages?

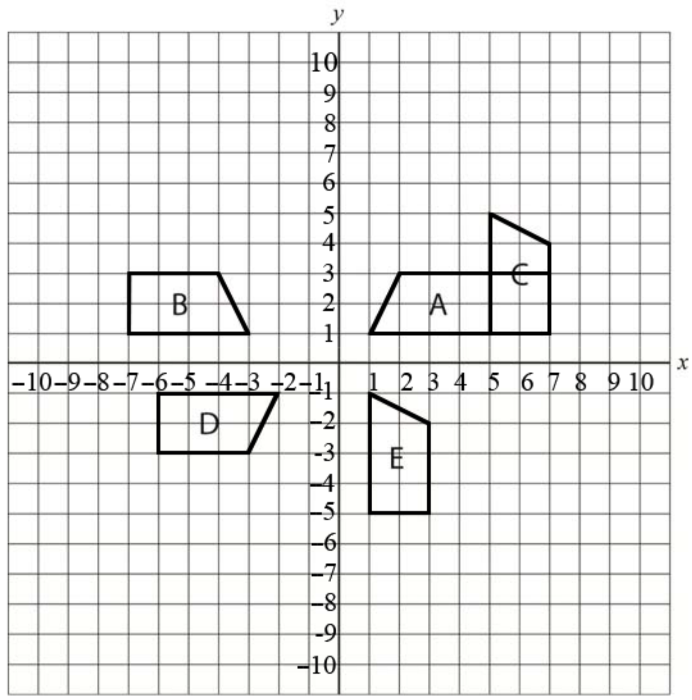
11 Jason left town A to travel to town B at 6.00 a.m. Town B is at least 900 km away from town A. He drove at an average speed of 90 km/h.
At 8.30 a.m., Simon left town A to travel to town B.
He drove at an average speed of 120 km/h.
At what time will Simon catch up with Jason?

12 Camille took 50 minutes to complete a journey. She travelled half the distance at a speed of 120 km/h and the other half at 80 km/h.
How far was her journey?

Core revision exercises: Shape and space

Worksheet 23: Transformations

1 Fully describe the single transformations that map shape A onto each of the other shapes in the diagram.



2 Draw the following transformations of triangle ABC.

a Reflect the triangle in the line $y = 0$.

b Reflect the triangle in the line $x = 0$.

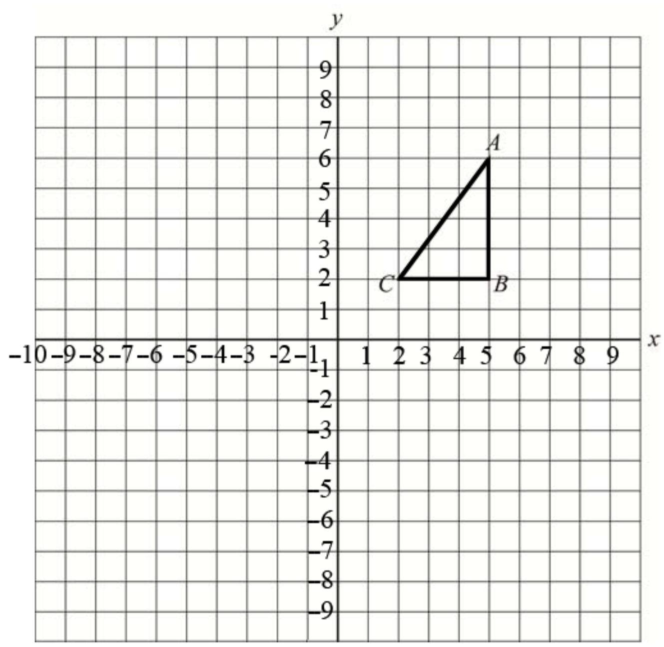
c Rotate the triangle 90° clockwise about the midpoint of the hypotenuse.

d Translate the triangle $\begin{pmatrix} -2 \\ -4 \end{pmatrix}$.

e Translate the triangle so that the coordinates of A' are $(-1, -1)$.

Write a column vector to describe this translation.

f Rotate the translated triangle (from part e) 180° about the point C'.



3 Given that $\mathbf{a} = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -3 \\ 5 \end{pmatrix}$, express $3\mathbf{a} - \mathbf{b}$ as a column vector.

4 Given that $\mathbf{a} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} -1 \\ 5 \end{pmatrix}$ and $\mathbf{c} = \begin{pmatrix} -4 \\ 2 \end{pmatrix}$, simplify:

a $\mathbf{a} + \mathbf{b} + \mathbf{c}$ **b** $2\mathbf{a} + 2\mathbf{b} + \mathbf{c}$ **c** $3\mathbf{a} + \mathbf{b} - \mathbf{c}$

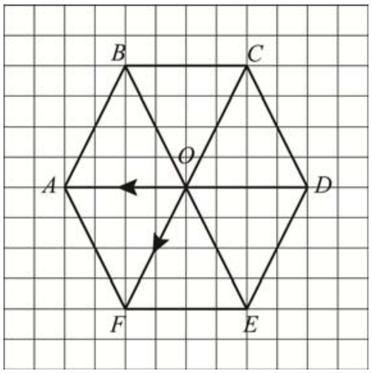
d $-\mathbf{a} - 2\mathbf{b} - \mathbf{c}$ **e** $3\mathbf{a} - 4\mathbf{b} + 3\mathbf{c}$

5 The diagram shows a regular hexagon $ABCDEF$ with centre O. $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OF} = \mathbf{b}$.

a Find in terms of **a** and **b**:

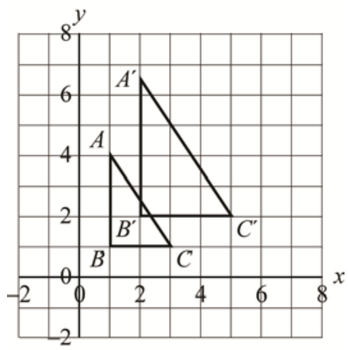
i AF **ii** OE

b Show that $AD = 2BC$.

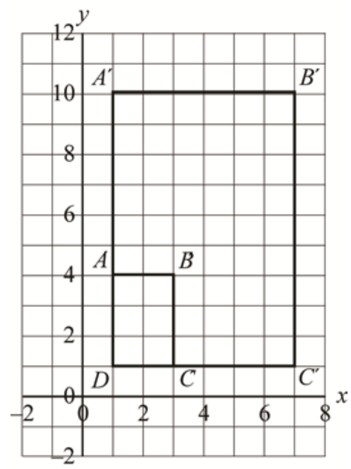


6 Describe these two enlargements.

a



b

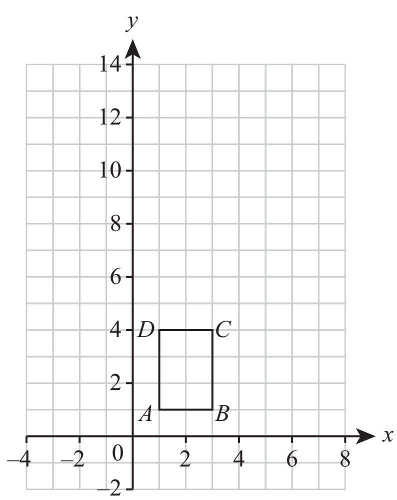


7 **a** Enlarge $ABCD$ by a scale factor of 2 using (0, 0) as the centre of enlargement. Label this $A'B'C'D'$.

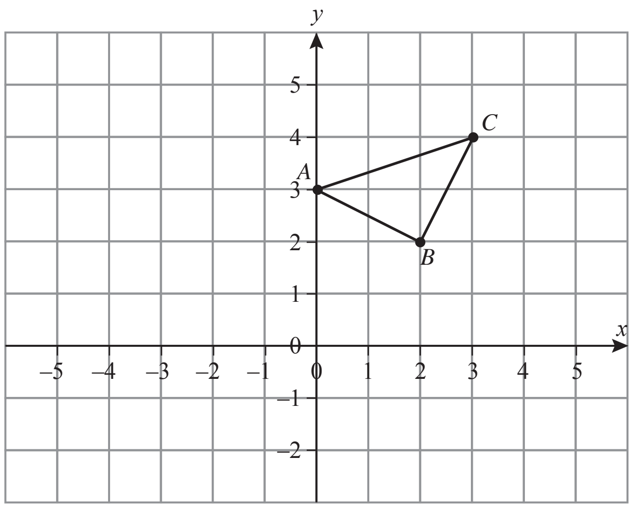
b On the same grid, enlarge $ABCD$ by a scale factor of 4 using (2, 1) as the centre of enlargement. Label this $A''B''C''D''$.

c Express the relationship between the length of AD and $A'D'$ as a ratio in simplest form.

d Express the relationship between $A''B''$ and AB as a ratio in simplest form.



8 Triangle ABC is drawn on the grid.



Draw the image $A'B'C'$ after an enlargement of scale factor 0.5 with centre $(-1, 0)$.

Core Revision exercises: Data handling

Worksheet 24: Probability using tree diagrams and Venn diagrams

1 A school cafeteria offers the following choices for lunch:

Starter	Main	Dessert
Soup	Curry	Lassi
Salad	Roti	Ice Cream
	Fish	
	Vegetarian	

Each student can choose a starter, a main course and a dessert.

a Draw a tree diagram to show all the possible choices for a three course lunch.

b What is the probability that a student will choose soup, curry and ice-cream?

2 Sally has a red and a blue pen, as does Soe. Mary has a red and a black pen. The teacher takes one pen from each girl at randam.

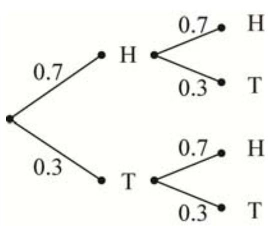
a Draw a tree diagram to show all the possible outcomes.

b What is the probability that the three pens chosen:

i are all red?

ii include only one red?

3 This tree diagram shows the probability of getting heads and tails when a biased coin is tossed twice.



Find:

a the probability of obtaining two heads

b the probability of obtaining two tails

c the probability of getting tails on the first toss and heads on the second toss

d the probability of getting only one tail

e the probability of getting at least one tail.

4 In a survey, 50 people were asked whether they read a newspaper or an online news site. 25 people read a newspaper, 31 read the news online and 15 people both read a newspaper and an online news site.

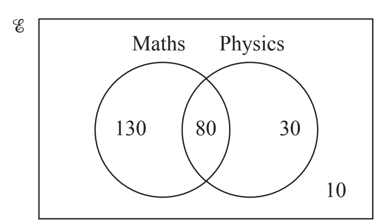
a Draw a Venn diagram to show this information.

b How many people in this sample read neither a newspaper nor an online site?

c What is the probability that a person in this sample reads only a newspaper?

d What is the probability that someone in this sample reads neither a newspaper nor an online site?

5 The Venn diagram shows data about the subjects taken by 250 students in an International School.



a How many of the students do not take maths or physics?

b What is the probability that a student chosen at random will take maths?

c Calculate the probability that a student chosen at random will take maths or physics.